

WILP, BITS Pilani

MACHINE LEARNING · COMPANION READER

Ensemble Methods & Text Classification

Committee Models, Bagging, Boosting, Multinomial Naïve Bayes, and the Connection to Logistic Regression

Session 12 · Read alongside the lecture slides · Every slide example worked out in full

Prepared for the Work Integrated Learning Programmes (WILP), BITS Pilani.

1 Ensemble Philosophy & Committee Models

Combining multiple diverse base classifiers reduces variance and produces superior predictive performance.

According to the **No Free Lunch Theorem**, no single learning algorithm outperforms all others across all tasks. **Ensemble Learning** constructs a committee of base models $C_1, C_2, \dots, C_T$ and aggregates their predictions.

The intuition

Wisdom of the crowd: A committee of diverse, independent decision-makers makes far fewer errors than any single individual expert.

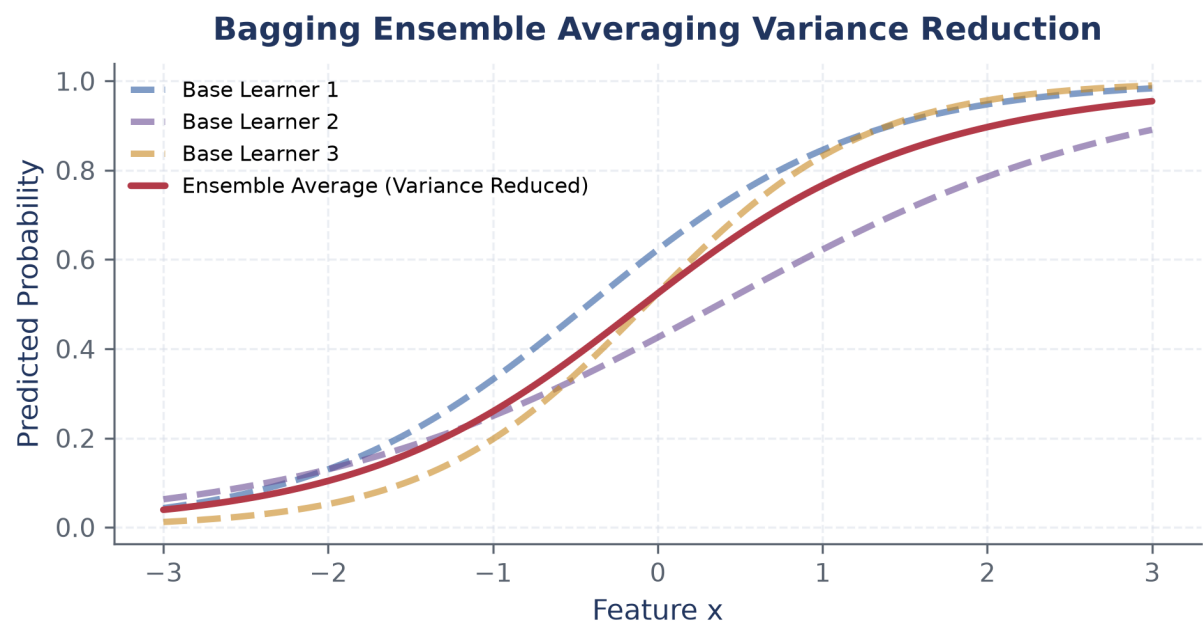


Figure 1: Bagging ensemble averaging reduces variance while maintaining low model bias.

2 Bagging (Bootstrap Aggregating)

Bagging creates diverse base models by training each on an independent bootstrap sample of the dataset.

Given a training set D of size N :

1. Generate T bootstrap samples $D_1, D_2, \dots, D_T$ by sampling N instances from D **with replacement**.
2. On average, each bootstrap sample contains $1 - (1 - 1/N)^N \approx 63.2\%$ of unique original instances, leaving 36.8% as Out-Of-Bag (OOB) validation data.
3. Train base classifier C_t on D_t .
4. Predict using majority voting (classification) or averaging (regression).

Worked example: 1D Bagging Ensemble Voting Across 8 Rounds

Consider 10 points $x \in \{0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0\}$. Decision stump base classifiers pick split thresholds across 8 bagging rounds:

Step 1. Round 1 (Split $x = 0.35$): Predicts +1 for $x \leq 0.35$, -1 for $x > 0.35$.

Step 2. Round 2 (Split $x = 0.70$): Predicts +1 for $x \leq 0.70$, +1 for $x > 0.70$.

Step 3. Round 6 (Split $x = 0.75$): Predicts -1 for $x \leq 0.75$, +1 for $x > 0.75$.

Step 4. Majority Vote Aggregation: Taking the mode vote across all 8 rounds smooths out local variance and produces a stable, solid decision boundary.

3 Multinomial Naïve Bayes for Text Classification

Multinomial Naïve Bayes models document term frequencies for text categorization.

In the **Bag of Words (BoW)** model, text order is ignored and word counts f_i represent the document. The posterior class score is:

$$P(c | d) \propto P(c) \prod_{i=1}^{|V|} P(w_i | c)^{f_i}, \quad P(w_i | c) = \frac{N_{c,w_i} + 1}{N_c + |V|}. \quad (1)$$

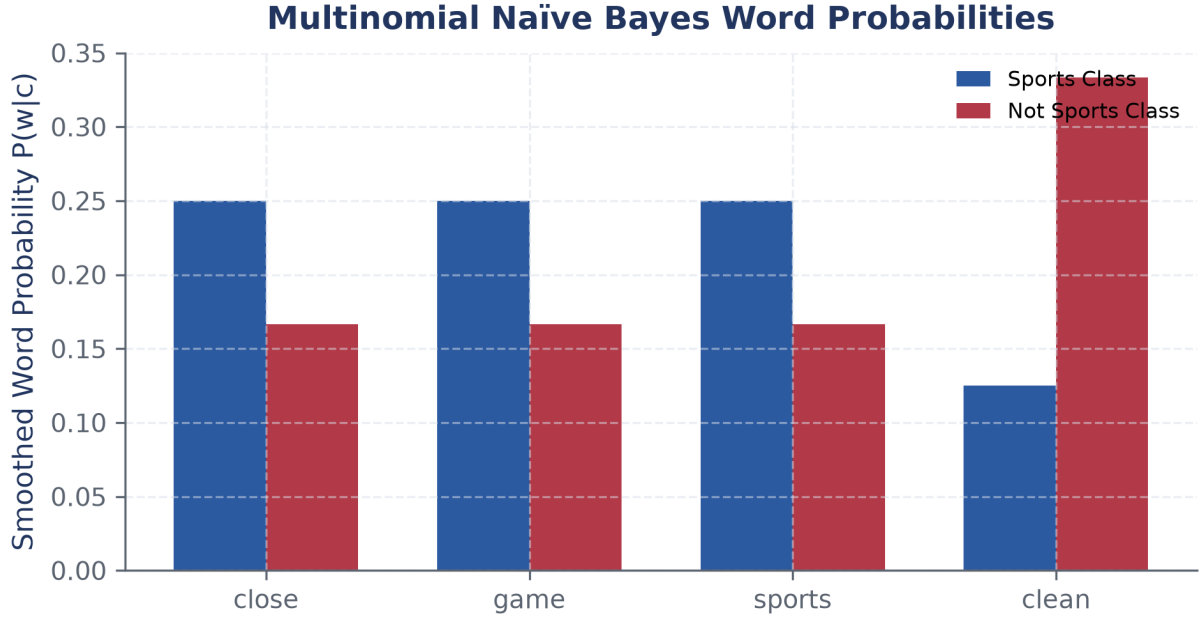


Figure 2: Multinomial word likelihood probabilities $P(w|c)$ for Sports vs Not Sports text classification.

Worked example: Sentence Tagging: "A very close game"

Classify sentence $d = \text{"A very close game"}$ into $c \in \{\text{Sports, Not Sports}\}$. Vocabulary size $|V| = 6$. Total words in Sports $N_{\text{Sports}} = 8$, in Not Sports $N_{\text{NotSports}} = 3$. Word count in Sports: $N_{\text{Sports, close}} = 1$, $N_{\text{Sports, game}} = 1$.

Step 1. Compute Smoothed Word Probabilities for Sports:

$$P(\text{close} | \text{Sports}) = \frac{1+1}{8+6} = \frac{2}{14} = \frac{1}{7}, \quad P(\text{game} | \text{Sports}) = \frac{1+1}{8+6} = \frac{2}{14} = \frac{1}{7}. \quad (2)$$

Step 2. Compute Smoothed Word Probabilities for Not Sports:

$$P(\text{close} | \text{NotSports}) = \frac{0+1}{3+6} = \frac{1}{9}, \quad P(\text{game} | \text{NotSports}) = \frac{0+1}{3+6} = \frac{1}{9}. \quad (3)$$

Step 3. Compare Class Scores:

$$P(\text{Sports} | d) \propto 0.5 \times \frac{1}{7} \times \frac{1}{7} = \frac{1}{98} \approx 0.0204, \quad (4)$$

$$P(\text{NotSports} | d) \propto 0.5 \times \frac{1}{9} \times \frac{1}{9} = \frac{1}{162} \approx 0.0123. \quad (5)$$

Since $0.0204 > 0.0123$, the sentence is correctly classified as **Sports!**

4 Mathematical Link: Gaussian Naïve Bayes & Logistic Regression

Gaussian Naïve Bayes with equal variances mathematically implies the Logistic Regression functional form.

For binary classification $Y \in \{0, 1\}$, assume features $X_i | Y = y \sim \mathcal{N}(\mu_{iy}, \sigma_i^2)$ with equal class variances $\sigma_{i0} = \sigma_{i1} = \sigma_i$.

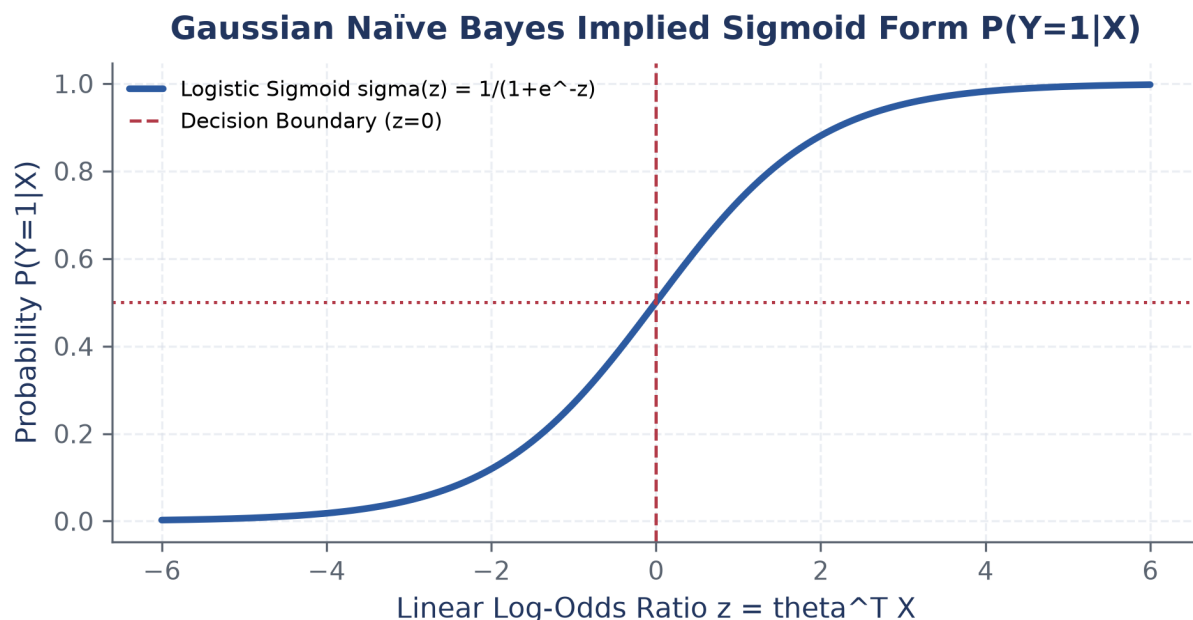


Figure 3: Logistic sigmoid curve derived from Gaussian Naïve Bayes log-odds ratio.

Applying Bayes' rule:

$$\begin{aligned}
 P(Y = 1 | \mathbf{X}) &= \frac{P(Y = 1)P(\mathbf{X} | Y = 1)}{P(Y = 1)P(\mathbf{X} | Y = 1) + P(Y = 0)P(\mathbf{X} | Y = 0)} \\
 &= \frac{1}{1 + \exp\left(-\ln \frac{P(Y=1)P(\mathbf{X}|Y=1)}{P(Y=0)P(\mathbf{X}|Y=0)}\right)}.
 \end{aligned} \tag{6}$$

Expanding the log-odds ratio with Gaussian densities:

$$\ln \frac{P(Y = 1)}{P(Y = 0)} + \sum_{i=1}^d \ln \frac{p(X_i | Y = 1)}{p(X_i | Y = 0)} = \theta_0 + \sum_{i=1}^d \theta_i X_i = \boldsymbol{\theta}^T \mathbf{X}, \tag{7}$$

where $\theta_i = \frac{\mu_{i1} - \mu_{i0}}{\sigma_i^2}$. So:

$$P(Y = 1 | \mathbf{X}) = \frac{1}{1 + e^{-\boldsymbol{\theta}^T \mathbf{X}}}. \tag{8}$$

Gaussian Naïve Bayes directly implies the exact sigmoid functional form of Logistic Regression!

Glossary & Symbols

Key Terms

Ensemble Learning

Combining predictions from multiple base models to improve stability and accuracy.

Bagging

Bootstrap Aggregating; sampling data with replacement to reduce variance.

Multinomial Naïve Bayes

Naïve Bayes model for discrete word frequency distributions.

Out-Of-Bag (OOB)

The $\approx 36.8\%$ of training samples excluded from a bootstrap sample.

Symbols at a Glance

Symbol	Meaning
C_t	Base classifier model t in ensemble
$ V $	Vocabulary size in text classification
$\theta^T \mathbf{X}$	Linear log-odds decision boundary score